Class Types in C#

In C#, classes form the foundation of object-oriented programming by defining the blueprint for creating objects. They encapsulate data and behavior and can be specialized into different types depending on the intended design and usage. Understanding the different class types helps developers design flexible, maintainable, and efficient applications.

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1. Regular Classes

A regular class is the most common type. It can contain fields, properties, methods, and events. Instances of such classes are created using constructors, and each instance has its own copy of non-static members.

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2. Abstract Classes

Abstract classes serve as base classes that cannot be instantiated directly. They define members that derived classes are expected to implement, ensuring a common contract. They can include both fully defined members and abstract members that lack implementation.

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3. Sealed Classes

Sealed classes prevent inheritance. They are used when a class's behavior should not be extended or modified further. This helps enforce strict behavior and can sometimes improve performance by allowing certain compiler optimizations.

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4. Static Classes

Static classes cannot be instantiated and can only contain static members. They are often used as utility containers for methods, constants, or configurations that apply to the program as a whole, rather than to individual objects.

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5. Partial Classes

Partial classes allow a class definition to be split across multiple files. At compile time, the parts are combined into a single definition. This approach improves code organization, especially in large projects or when working with generated code.

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6. Nested Classes

Nested classes are defined within another class. They are useful for logically grouping types that are only relevant in the context of their containing class. Nesting also allows closer control of access levels.

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7. Generic Classes

Generic classes use type parameters to define members and operations in a type-safe way without committing to a specific data type. They provide reusability and flexibility while maintaining compile-time type safety.

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8. Partial Abstract or Generic Combinations

Classes can also combine features of different types, such as being both partial and abstract or generic and nested. These combinations are valid and allow even more control over structure and behavior.